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## Series 10: Abstract Interpretation II/III

### Methods

#### Analysis of Order Structures

- **Verification of lattice properties:** Determine if a given poset is a lattice, a complete lattice, or a semi-complete lattice by checking the existence of infimum and supremum for any pair (lattice) and for any subset (complete).
- **Construction of order on abstract domains:** Define an order that captures the notion of “precision” between abstract elements (e.g., inclusion for intervals).
- **Cartesian product of lattices:** Show that the product of two complete lattices, equipped with the product order, is itself a complete lattice.

#### Analysis of Functions on Lattices

- **Monotonicity and Scott-continuity:** Prove that a function is monotonic and/or Scott-continuous. Scott-continuity guarantees preservation of suprema of increasing chains.
- **Composition of continuous functions:** Show that the composition of two Scott-continuous functions is Scott-continuous.
- **Fixed-point calculation:** Apply Kleene’s theorem to compute the least fixed point (lfp) of a continuous function, often via iteration starting from  $\perp$ .

- **Comparison of fixed points:** If  $f(x) \leq g(x)$  for all  $x$ , then  $\text{lfp}(f) \leq \text{lfp}(g)$ .

## Galois Connections

- **Definition of functions  $\alpha$  (abstraction) and  $\gamma$  (concretization)** between a concrete domain and an abstract domain.
- **Proof of fundamental properties:** Verify that  $(\alpha, \gamma)$  indeed form a Galois connection, i.e.,  $\alpha(X) \subseteq i \Leftrightarrow X \subseteq \gamma(i)$ .
- **Derived properties:** Show that  $\alpha$  and  $\gamma$  are monotonic, that  $x \leq \gamma(\alpha(x))$ , and that  $\alpha(\gamma(y)) \leq y$ .

## Abstract Interpretation of Programs

- **Construction of the equation system:** From a CFG (Control Flow Graph), write the fixed-point equations for each program point.
- **Iterative resolution:** Apply Kleene iteration on an abstract domain (e.g., sign lattice, interval lattice) until convergence.
- **Condition handling:** Define the “filtering” operator  $(\cap[\textit{condition}])$  to refine information at a guard (e.g.,  $x \geq y$  in the interval domain).

## Reminder

- **Complete lattice:** Every subset has an infimum and a supremum.
- **Scott-continuity:**  $f(\sup_i u_i) = \sup_i f(u_i)$  for any increasing chain  $(u_i)$ .
- **Kleene’s theorem:** In a complete lattice, a Scott-continuous function has a least fixed point given by  $\sup_{n \geq 0} f^n(\perp)$ .
- **Galois connection:** Pair  $(\alpha : \mathcal{C} \rightarrow \mathcal{A}, \gamma : \mathcal{A} \rightarrow \mathcal{C})$  satisfying  $\forall X \in \mathcal{C}, \forall i \in \mathcal{A}, \alpha(X) \subseteq i \Leftrightarrow X \subseteq \gamma(i)$ .
- **Product order:**  $(a, b) \leq (c, d) \Leftrightarrow a \leq c \wedge b \leq d$ . Infimum and supremum are computed componentwise.

## Illustrative Examples

### Example 1: Verification that a poset is a complete lattice

**Context:** The exercise asks whether  $(\mathbb{N}, \leq)$  is a (complete) lattice.

**Application:**

- It is a lattice because for any pair  $\{a, b\}$ ,  $\inf(a, b) = \min(a, b)$  and  $\sup(a, b) = \max(a, b)$  exist in  $\mathbb{N}$ .

- It is **not complete** because the subset  $X = \mathbb{N}$  has no supremum in  $\mathbb{N}$  ( $\sup(\mathbb{N}) = +\infty \notin \mathbb{N}$ ).
- It is a **semi-complete lattice** because every subset  $A \subseteq \mathbb{N}$  has an infimum (its minimum).
- However,  $(\mathbb{N} \cup \{\infty\}, \leq)$  is a complete lattice.

### Example 2: Definition of an order and inf/sup calculation on the interval domain

**Context:** For the set of intervals  $\mathcal{I}$ , define an order representing loss of precision.

**Application:**

- Define the order by **reverse inclusion** of value sets:  $(a, c) \leq (b, d) \Leftrightarrow [b, d] \supseteq [a, c] \Leftrightarrow (b \leq a \wedge c \leq d)$ .
- The infimum corresponds to the **intersection** of value sets:

$$\inf((a, c), (b, d)) = (\max(a, b), \min(c, d))$$

(which gives the smallest interval contained in both).

- The supremum corresponds to the **union** (the smallest over-approximation):

$$\sup((a, c), (b, d)) = (\min(a, b), \max(c, d))$$

- Prove that  $(\mathcal{I}, \leq)$  is a complete lattice using this correspondence with set inclusion.

### Example 3: Proof that a Scott-continuous function is monotonic

**Context:** Let  $f : E \rightarrow F$  be Scott-continuous, with  $E, F$  complete lattices.

**Application:** Let  $x, y \in E$  such that  $x \leq y$ .

1. Construct the increasing chain  $(u_i)$ :  $u_0 = x$ , and  $u_i = y$  for all  $i \geq 1$ .
2. By construction,  $\sup_i u_i = y$ .
3. By Scott-continuity:  $f(y) = f(\sup_i u_i) = \sup_i f(u_i)$ .
4. In particular,  $f(u_0) = f(x) \leq \sup_i f(u_i) = f(y)$ .
5. Therefore  $f(x) \leq f(y)$ , which proves monotonicity.

### Example 4: Construction of a Galois Connection between $\mathcal{P}(\mathbb{Z})$ and $\mathcal{I}$

**Context:** Define  $\alpha$  and  $\gamma$  between the power set of  $\mathbb{Z}$  (concrete) and the interval lattice  $\mathcal{I}$  (abstract).

**Application:**

- **Abstraction function  $\alpha$ :**

$$\alpha : \mathcal{P}(\mathbb{Z}) \rightarrow \mathcal{I}, \quad \alpha(X) = [\inf X, \sup X]$$

(with  $\inf \emptyset = +\infty$ ,  $\sup \emptyset = -\infty$  by convention).

- **Concretization function  $\gamma$ :**

$$\gamma : \mathcal{I} \rightarrow \mathcal{P}(\mathbb{Z}), \quad \gamma([a, b]) = \{z \in \mathbb{Z} \mid a \leq z \leq b\}$$

- **Proof of the connection:** Must show  $\forall X \subseteq \mathbb{Z}, \forall i \in \mathcal{I}$ :

$$\alpha(X) \subseteq i \Leftrightarrow X \subseteq \gamma(i)$$

1. ( $\Rightarrow$ ) If  $[\inf X, \sup X] \subseteq i$ , then every  $x \in X$  satisfies  $x \in \gamma(i)$ .
2. ( $\Leftarrow$ ) If  $X \subseteq \gamma(i)$ , then  $\inf X \geq a$  and  $\sup X \leq b$ , so  $[\inf X, \sup X] \subseteq [a, b] = i$ .

### Example 5: Program Analysis with the Interval Lattice

**Context:** Analyze the following program with the interval domain:

```
y := 3; x := 4;
while (x >= y) do x := x - 4; done;
```

#### Application:

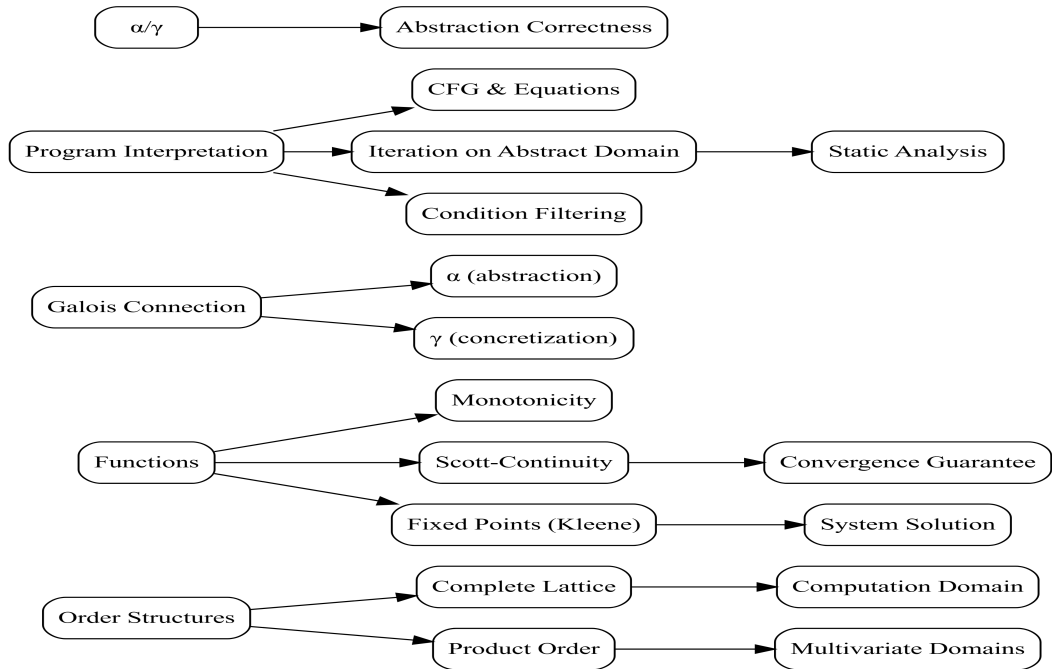
##### 1. CFG and equations:

- Entry:  $v_0 = (x : \top, y : \top)$
- $v_1 = (x : [4, 4], y : [3, 3])$  after assignments.
- Loop (point  $v_2$ ):  $v_2 = v_1 \sqcup F(v_2)$ , where  $F$  is the transfer function of the body.
- Loop body:  $F((x, y)) = (x - 4, y) \cap [x \geq y]$ .

##### 2. Kleene iteration:

- $v_2^{(0)} = \perp = (x : \perp, y : \perp)$
- $v_2^{(1)} = v_1 \sqcup F(\perp) = (x : [4, 4], y : [3, 3])$
- Compute  $F(v_2^{(1)})$ :  $(x : [4, 4] - 4, y : [3, 3]) = (x : [0, 0], y : [3, 3])$ . The condition  $x \geq y$  becomes  $[0, 0] \geq [3, 3]$  which is false, so the result is  $\perp$ .
- $v_2^{(2)} = v_1 \sqcup \perp = v_2^{(1)}$ . Fixed point reached.

3. **Result:** At loop exit,  $x = [4, 4]$  and  $y = [3, 3]$ . The loop does not execute.



## Synthesis

This series highlights the mathematical foundations of abstract interpretation: **complete lattices** serve as computation domains, **Scott-continuity** ensures iteration convergence, and **Galois connections** guarantee abstraction correctness.

The core method combines modeling a program by a **fixed-point equation system** and its iterative resolution on an abstract domain (signs, intervals).

These techniques allow automatically inferring properties about programs (variable bounds, signs) without executing them.